

DBMS & Relational Algebra Course Notes

Chapter 1: Relational Model Foundations

The relational model represents data as mathematical relations (tables). Each row represents a tuple a
Keys:

- Primary Key: Uniquely identifies a tuple in a table.
- Foreign Key: Enforces referential integrity between two related tables.
- Candidate Key: Minimal superkey with no redundant attributes.

Chapter 2: Normalization and Functional Dependencies

Normalization reduces data redundancy through decomposition.

- 1NF: Atomic column values.
- 2NF: No partial dependency on candidate keys.
- 3NF: No transitive dependency on non-prime attributes.
- BCNF: Determinant of every non-trivial functional dependency must be a superkey.